

# The Gradient of the Logistic Function

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## 1 What does a Logistic Function do?

Take one row of observations. Let there be  $d$  observations and a classification  $y$  where  $y \in \{0, 1\}$ . You can think of  $y$  as yes or no, or true or false. Let each of the  $d$  observations we represented by a variable  $x_j$ . Where each  $x_j$  is the z-score for observation (the standard deviations from the mean).

Our goal is simply take these  $x_j$  and predict  $y$  as best as possible. So intuitively we can take a single observation  $x_2$  say and whenever its  $> 1$  say we let our prediction by 1 for  $y$ . But this doesn't take into account all our observations. So let sum them, let  $z = x_1 + x_2 + \dots + x_d + b$ . We introduced a  $b$  which is a bias for the guess we want. You can imagine  $b$  as if all  $x_j$  are 0 then what should our guess be.

Now you might think now that some observations may be more important than others or interact with  $y$  in different ways. For example your length of credit history might be more important than your hair color in predicting if you will repay or not.

So assign a weight  $w_j$  to each observation  $x_j$ . So we have now  $z = w_1x_1 + w_2x_2 + \dots + w_dx_d + b$ . (Make these equations).

But how do we get these weights? We need to say based on these current weights how confident are we that  $y = 1$  ie given this value of  $z$  how confident are we that this observation as  $y = 1$ . Well  $z$  is arbitrary,  $w_1$  can be as big as we want  $z$  can be 100,  $-0.5$  so on; what does that mean?

Take a function  $\frac{1}{1+e^{-z}}$  and let this be how predicted probability that  $y = 1$ . Specifically let  $p = \sigma(z) = \frac{1}{1+e^{-z}}$ . Then observe that for every  $\epsilon > 0$  there exists  $N \in \mathbb{N}$  such that whenever  $z \geq N$   $|\frac{1}{1+e^{-z}} - 1| < \epsilon$ , specifically choose

$z > \log(\frac{1}{\epsilon})$ . To see this  $|\frac{1}{1+e^{-z}-1}| = |\frac{1}{1+e^{-z}} - \frac{1+e^{-z}}{1+e^{-z}}| = |\frac{-e^{-z}}{1+e^{-z}}| = \frac{e^{-z}}{1+e^{-z}}$  since  $e^x > 0$  always we have  $1+e^{-z} > 1$  giving  $\frac{e^{-z}}{1+e^{-z}} < e^{-z}$  now  $e^{-z} < \epsilon = -z < \log(\epsilon) = z > -\log(\epsilon) = z > \log(\frac{1}{\epsilon})$ . Meaning whenever  $z > N$  where  $N > \log(\frac{1}{\epsilon})$  we have  $|\frac{1}{1+e^{-z}-1}| < \epsilon$  giving  $p- > 1asz- > \inf$ .

For the other direction  $|\frac{1}{1+e^{-z}-0}| < \epsilon$  where  $|\frac{1}{1+e^{-z}-0}| = \frac{1}{1+e^{-z}}$  since  $e^x > 0$  always. Then  $1 < \epsilon(1+e^{-z})$  and  $\frac{1-\epsilon}{\epsilon} < e^{-z}$ , taking log of both sides giving  $\log(\frac{1-\epsilon}{\epsilon}) < -z$ , then multiplying by  $-1$  giving  $-\log(\frac{1-\epsilon}{\epsilon}) > z$ . Meaning  $|\frac{1}{1+e^{-z}-0}| < \epsilon$  for large  $z$  giving  $p- > 0asz- > -\inf$ .

Now that we have proved  $p \in (0, 1)$  and we have defined it as the probability that  $y = 1$  for our model we can define the loss that we are looking to minimize. Intuitively we want our probability of  $y = 1$  distance from the true value. But an absolute difference doesn't account for the confidence difference we between a precision of .99 and .999 which are 0.01 and 0.001 away from 1 but are dramatically different in confidence. Therefore we define the loss using logs. Specifically we let the loss  $= -\log(p)$  if  $y = 1$  and  $-\log(1-p)$  if  $y = 0$ . In closed form our loss looks like  $L = -[y \log(p) + (1-y) \log(1-p)]$ .

As stated before we want to minimize our loss but we want to tune our arbitrary weights. So how can we do this. We want to tune our weights as optimally as possible. Therefore we use the gradient specifically  $\frac{\partial L}{\partial w_j}$ . But we defined  $L$  in terms of  $y$  and  $p$  so we must chain through to get our derivative.

Observe by the chain rule  $\frac{\partial L}{\partial w_j} = \frac{\partial L}{\partial p} \frac{\partial p}{\partial z} \frac{\partial z}{\partial w_j}$

Moving from right to left we have  $\frac{\partial z}{\partial w_j} = x_j$  since taking the partial derivative leaves all other weights as constants

Next  $\frac{\partial p}{\partial z} = \sigma'(z) = -1(1+e^{-z})^{-2}e^{-z} = \frac{e^{-z}}{(1+e^{-z})^2}$ . Observe that  $\frac{e^{-z}}{(1+e^{-z})^2} = \frac{e^{-z}}{(1+e^{-z})} \frac{1}{(1+e^{-z})}$  where the right side is again  $\sigma(z)$  and the left side is  $1 - \sigma(z)$  since  $1 - \sigma(z) = 1 - \frac{1}{1+e^{-z}} = \frac{1+e^{-z}}{1+e^{-z}} - \frac{1}{1+e^{-z}} = \frac{e^{-z}}{(1+e^{-z})}$ . Giving  $p' = p(1-p)$

Finally  $\frac{\partial L}{\partial p} = \frac{-y}{p} + \frac{1-y}{1-p}$

Now chaining all three together gives  $(\frac{-y}{p} + \frac{1-y}{1-p})p(1-p)x_j = (-y(1-p) + (1-y)p)x_j = (-y + py + p - py)x_j = (p - y)x_j$  For  $b$  simply observe  $\frac{\partial z}{\partial b} = 1$  and therefore the gradient is  $(p - y)$  there.

We now have the optimal unit of movement of each  $w_j$  in order to most efficiently decrease our loss.

In general we have  $d$  columns of observations and  $n$  rows of data. We can then simply think of each variable described above as a vector. Where  $w$  is the weight vector in which we add  $b$ .  $p$  is the vector of probabilities one for each row  $p_i$ .  $y$  is the vector of observations for rows  $i$  where  $y_i$  is the value 0 or 1 described throughout. To get our final gradient we take Loss as the average loss of the rows. So  $L = \frac{1}{n} \sum_{i=1}^n L_i$  where  $L_i = (p_i - y_i)x_i$  for each observation  $j$ . Then to find the partial derivative of the gradient we have  $\frac{\partial L}{\partial w_j} = \frac{1}{n} \sum_{i=1}^n \frac{\partial L_i}{\partial w_{ij}}$  since the partial derivative of the mean is the mean of the partial derivatives.

In general at optimum we have  $\sum_{i=1}^n p_i = \sum_{i=1}^n y_i$  and our logistic regression produces its most optimal predictions based on the data given which we can further use to predict future data.